AI-BASED EARLY CHILDHOOD EMOTIONAL INTELLIGENCE MONITORING PLATFORM IN URBAN AREAS

Project Id: 25-26J-088

Project Proposal Report

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Sri Lanka

August 2025

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DECLARATION

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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27/08/2025

Signature of the supervisor

Date

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27/08/2025

Signature of the co-supervisor

Date

Mrs. Lokesha Weerasinghe

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ABSTRACT

Language development during early childhood is a critical component of cognitive, social, and emotional growth, particularly for second-language learners. In Sri Lanka, many children aged 4–6 acquire English as a second language, yet face challenges in grammar accuracy, pronunciation, and conversational fluency. Existing language assessment tools are often designed for older native speakers, lack personalization, and fail to provide actionable, explainable feedback for educators and parents. This research proposes the development of an AI-powered conversational voice assistant that engages children in short, interactive English dialogues while assessing their grammatical proficiency. The system integrates a fine-tuned child-friendly speech recognition model to ensure accurate transcription for young English as a second language learners, and a deep learning grammar correction model trained on locally relevant sentence structures. Using reinforcement learning, the voice assistant dynamically adapts conversation complexity based on the child's grammar performance, creating a personalized learning experience. Post-session, the system provides explainable AI insights via a teacher–parent dashboard, including progress tracking over time and specific recommendations to improve grammar and conversational skills. By linking language development with Emotional Intelligence pillars, this research aims to deliver an innovative, culturally relevant solution for early childhood English education in Sri Lanka, bridging gaps in current language assessment methodologies.

Keywords: Language development, Early childhood, English as a second language learners, Conversational voice assistant, Emotional Intelligence.

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LIST OF ABBREVIATIONS

ESL - English as a Second Language

EFL – English as a Foreign Language

AI – Artificial Intelligence

EI – Emotional Intelligence

CA – Conversational Agents

RL - Reinforcement Learning

XAI - Explainable AI

ASR - Automatic Speech Recognition

GPU - Graphics Processing Unit

GED - Grammatical Error Detection

GEC - Grammatical Error Correction

1. INTRODUCTION

Emotional Intelligence, usually described as the ability to recognize, understand, manage, and use emotions effectively in oneself and others [1]. By developing emotional intelligence, people learn more about themselves and learn to cooperate with those around them [2]. For early childhood, emotional intelligence is crucial because lower emotional intelligence impacts clear thinking process and the education of the preschoolers [3].

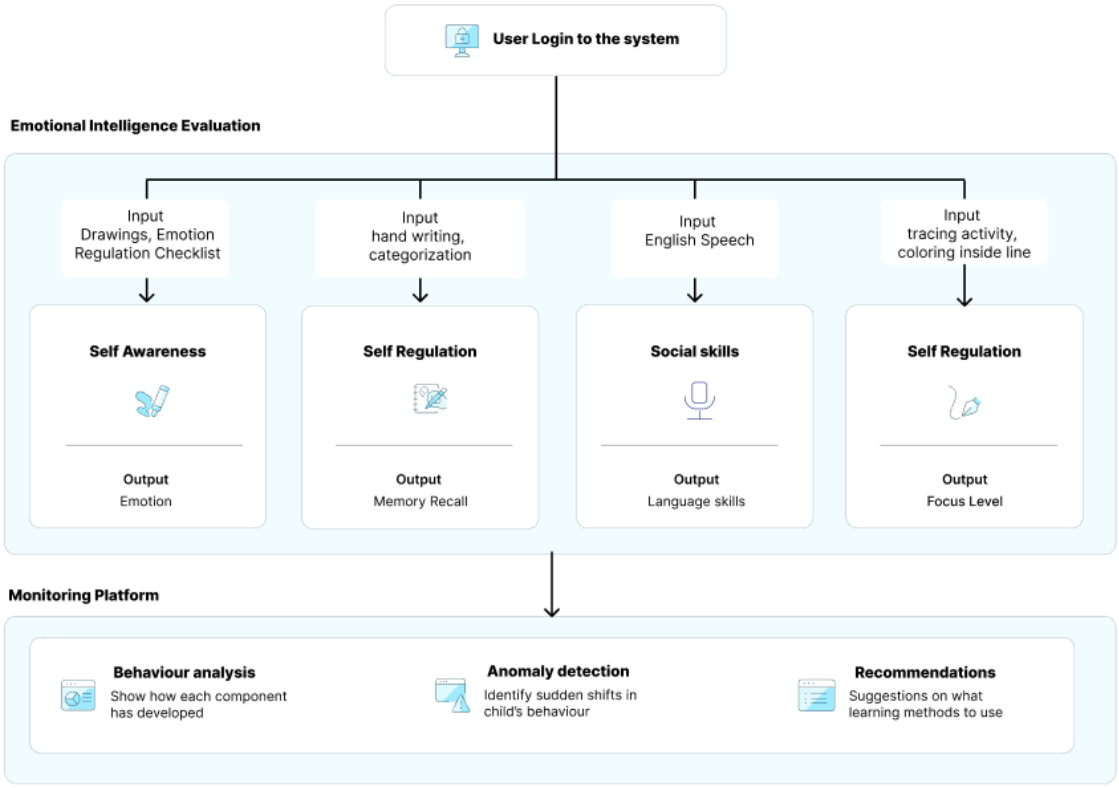


Figure 1.1: System Overview Diagram

Language acquisition during early childhood plays a pivotal role in a child's overall development, influencing not only communication skills but also cognitive, emotional, and social growth. According to developmental psychology research, the ages of 4–6 represent a critical period in which children rapidly expand their vocabulary, improve grammatical structures, and refine pronunciation. These formative years are particularly

important for second-language learners, as early exposure and practice in a supportive environment can significantly enhance long-term language proficiency [4], [5].

In Sri Lanka, English is often taught as a second language in preschool and primary education, serving as a bridge to global communication and academic opportunities. However, for many young learners, especially those in non-English-speaking households, mastering grammar and conversational fluency remains a challenge. Pronunciation influenced by the native language, limited opportunities for immersive conversation, and inadequate tailored feedback contribute to slow progress in second-language acquisition [6].

Artificial Intelligence (AI) technologies such as speech recognition, natural language processing, and conversational agents have emerged as promising tools in education, offering automated assessment, personalized learning, and real-time feedback. Yet, most AI-based language learning systems are designed for older learners or native English speakers, and few address the unique needs of young second-language learners in Sri Lanka. Furthermore, existing tools rarely integrate multiple capabilities such as accurate child-specific speech recognition, grammar detection and correction, adaptive conversation strategies, and explainable feedback into one cohesive platform that supports both children and educators [7].

This research aims to address these gaps by developing a conversational voice assistant specifically designed for Sri Lankan children aged 4–6 who are learning English as a second language. By combining fine-tuned speech recognition, deep learning-based grammar correction, reinforcement learning for adaptive dialogue, and explainable AI for progress tracking, the system will provide a culturally relevant, interactive, and educationally effective solution for early childhood language development.

1.1. Background & Literature Survey

Sulistyawati (2018) [8] examines how Emotional Intelligence (EI) influences students' English speaking abilities. The study focused on 12th-grade students in East Jakarta, utilizing questionnaires, objective assessments, and oral tests. Statistical analysis revealed a strong and significant positive relationship between EI and speaking performance students with higher EI demonstrated markedly better speaking competence. The findings underscore the critical role of emotional competencies such as self-awareness, self-regulation, empathy, and interpersonal skills in facilitating effective oral communication and managing speaking-related anxieties.

This study [9] investigated how emotional intelligence (EI) influences the development of oral English communication skills among Ukrainian students of ESL. The researchers measured EI using the Mayer-Salovey-Caruso Emotional Intelligence Test (MSCEIT) and assessed oral communication skills with a specially developed test on a 100-point scale. They also explored whether the deliberate integration of content-related emotions into speaking and listening activities could boost both EI and oral proficiency. Conducted over the 2018–2019 academic year with 124 undergraduate students of International Relations, the experimental group exposed to emotionally infused learning demonstrated significantly greater improvements in both emotional intelligence and oral communication skills compared to the control group, which saw only marginal gains. These results suggest a positive correlation between emotional engagement in content and enhancement in both emotional awareness and spoken English proficiency.

The study titled "Language Skills as Predictors of Social Skills and Behaviors in Preschool Children" by Nichole Mulvey and Lyndsay Jenkins (published April 10, 2020) [10] examined whether preschool children's language abilities measured through standardized assessments could predict teacher-reported social skills (namely empathy and assertion) and social behaviors (prosocial actions, aggression, and victimization). Involving 98 preschoolers and input from nine classroom teachers, correlational analyses revealed that stronger language skills were associated with better social skills and greater prosocial behavior, and conversely, with lower levels of victimization.

1.1.1. Background Survey

Our team carried out an online survey to find out how important it is to keep track of emotional intelligence in early childhood. This survey was conducted by sending the online form to parents and the teachers in urban areas in Western province. The survey results will help to build the application in a more successful manner. The Google form was filled by 87 individuals. Below are the results of the form.

Figure 1.2: Survey result 1

Figure 1.3: Survey result 2

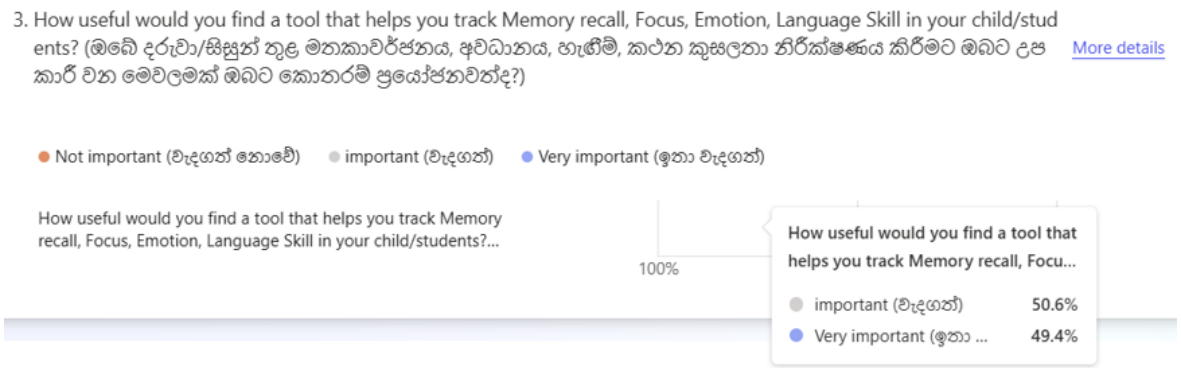


Figure 1.4: Survey result 3

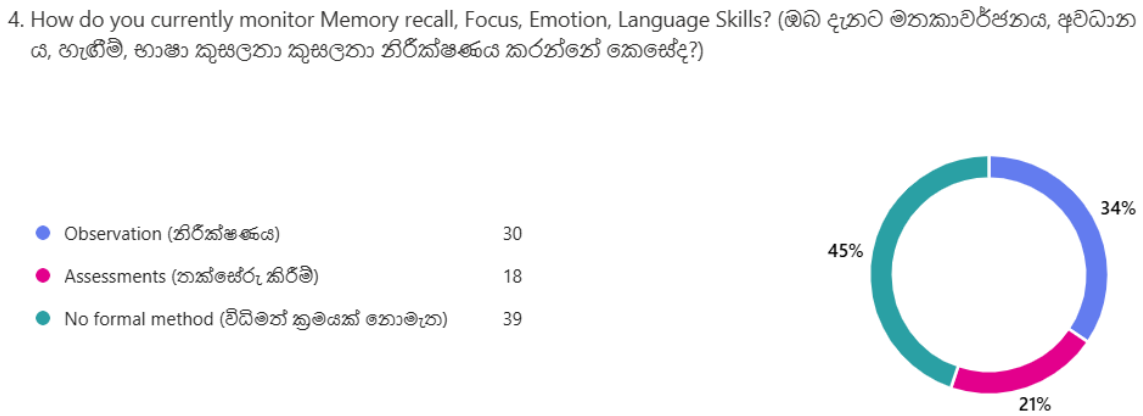


Figure 1.5: Survey result 4

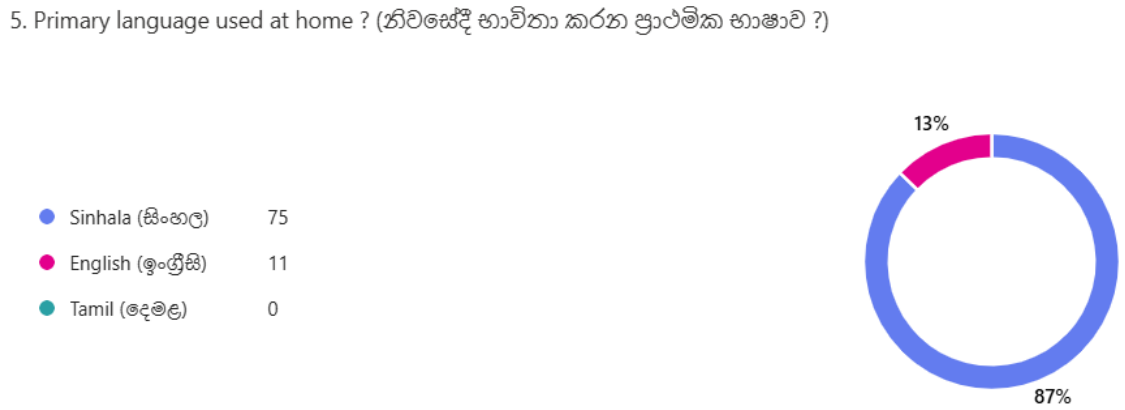


Figure 1.6: Survey result 5

6. Do you use English as the secondary language? (ඔබ ද්විතීයික භාෂාව ලෙස ඉංග්‍රීසි භාවිතා කරනවාද?)

Figure 1.7: Survey result 6

7. What activities does your child do most? (ඔබේ දරුවා වැඩිපුරම සිදුකරන ක්‍රියාකාරකම් මොනවාද?)

Figure 1.8: Survey result 7

8. How often does your child write letters at home ? (ඔබේ දරුවා නිවසේදී කොපමණ වාරයක් අකුරු ලියනවාද ?)

Figure 1.9: Survey result 8

9. Do you think an AI tool will be good for this? (මේ සඳහා AI මෙවලමක් හොඳ යැයි ඔබ සිතනවාද?)

Figure 1.10: Survey Result 9

10. How often would you use this tool? (ඔබ මෙම මෙවලම කොපමණ වාරයක් භාවිතා කරයිද?)

Figure 1.11: Survey result 10

11. How important is it to monitor children's emotional well-being? (දරුවන්ගේ චිත්තවේගීය යහපැවැත්ම නිරීක්ෂණ ය කිරීම කොතරම් වැදගත්ද?)

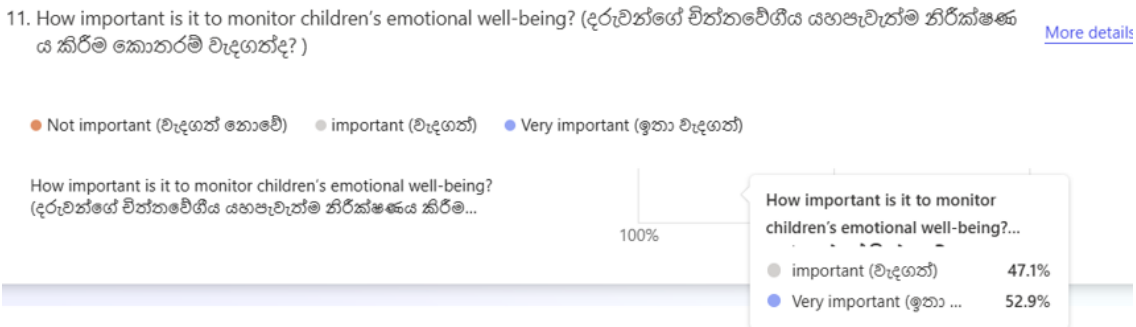


Figure 1.12: Survey result 11

This survey was filled mostly by parents whose children, go to Sinhala medium preschools, mostly speaking in Sinhala and used English as the secondary language. Most of the responders agreed, a tool that can track memory recall, focus, emotion and language skill will be useful. Currently no formal method is being used for these tasks. 85% of the people agreed that an AI tool would be helpful. Out of 87 people who filled the form, 64 responded that this should be assessed in a weekly basis. Overall, every single respondent agreed that monitoring a child's emotional well-being is a crucial tack.

1.1.2. Literature Survey

The study proposed by Xiao et al. (2023) conducted a scoping review of 23 studies sourced from APA PsycINFO, ERIC, and ProQuest Dissertations & Theses, exploring the use of conversational agents, voice-based systems like Siri, Alexa, or custom-built dialogue agents to support language learning [11]. The review identified three primary application modes: general communication practice where learners engage in spontaneous dialogue with CAs, task-based interactions involving specific activities mediated by the agent and structured, pre-programmed dialogues, designed around set topics or linguistic patterns [12]. Findings across these studies generally show that learners find CAs engaging, motivational, and supportive of additional practice outside classroom settings, with some evidence suggesting CAs can serve as diagnostic tools [12].

Reinforcement Learning has been successfully applied in dialogue systems to optimize conversational policies, primarily within adult-oriented, task-based, or open-domain settings. Surveys such as Kwan et al. (2023) [13] highlight RL's potential for adaptive conversations but also note its predominant focus on coherence, engagement, or task completion as reward signals rather than direct language learning outcomes. Educational chatbots for language learning have largely been rule-based or scripted, providing limited personalization and relying on predefined dialogue flows [14], [15]. While Grammar Error Correction (GEC) systems have leveraged RL to optimize correction quality in static text

[16], [17], these approaches do not operate in interactive, turn-by-turn tutoring contexts especially not with spoken input from children. Studies on language-learning conversational agents report improved learner engagement and confidence [18], [19] but often lack dynamic policy adaptation driven by real-time performance metrics such as grammar accuracy. Additionally, RL applications in education remain constrained by the absence of outcome-driven reward designs and validated classroom deployments [20], creating opportunities for innovative, learner-centered implementations.

Recent advances in Explainable AI shown promising results in enhancing transparency and interpretability within educational language assessments. For instance, Ribeiro-Flucht, Chen, and Meurers (2024) [21] introduced a comprehensive XAI approach that leverages 465 linguistic complexity measures to model proficiency and readability in Portuguese. By linking these theoretical measures with machine learning classification via Shapley Additive Explanations (SHAP), their system achieved state-of-the-art accuracies of 0.70 for proficiency classification and 0.84 for readability prediction. Importantly, their work demonstrates that interpretable features such as lexical complexity tied to proficiency can substantially enhance educational AI systems by making model decisions clear and explainable. This underscores the value of explainable mechanisms to support learner interpretation and trust in AI-generated feedback.

The proposed system in "Enhancing Grammatical Error Detection using BERT with Cleaned Lang-8 Dataset" [22] focuses on improving grammatical error detection (GED) performance by combining a rigorous data-cleaning pipeline with transformer-based language models. Starting from 2.35 million sentence pairs in the Lang-8 learner corpus, the authors implemented multi-stage preprocessing including duplication removal, normalization, punctuation standardization, contraction expansion, and Levenshtein distance filtering to produce a high-quality subset of 200,000 error–correction pairs. This cleaned dataset was used to fine-tune four transformer variants (BERT-base, BERT-large, RoBERTa-base, RoBERTa-large) for binary GED tasks, where each token is classified as correct or erroneous. Training was performed using AdamW optimization, a dropout rate

of 0.65, and selective layer freezing guided by the WeightWatcher tool to balance learning efficiency and performance. Experimental results demonstrated that BERT-base, when trained on 180k high-quality sentence pairs, achieved an F1 score of 0.91 and outperformed larger models, highlighting that data quality and targeted fine-tuning can yield superior GED accuracy even with smaller, computationally efficient architectures.

The research paper "Balarila: Deep Learning for Semantic Grammar Error Correction in Low-Resource Settings" [23] introduces a deep learning-based grammatical error correction (GEC) system tailored for the Filipino language, addressing both syntactic and semantic errors. The system, named Balarila, is inspired by the GECToR approach and utilizes transformer-based models such as Tagalog BERT and RoBERTa. Due to the lack of an existing annotated dataset, the authors developed an automated error generation pipeline that created synthetic datasets of error-free and error-filled Filipino sentences sourced from various online news platforms. These datasets were then used to fine-tune the models in two stages. The evaluation metrics included precision, recall, and F0.5 scores for GEC, and a multi-class confusion matrix for grammatical error detection (GED). The top-performing model, RoBERTa Tagalog Large, achieved an F0.5 score of 70.75, while the RoBERTa Tagalog Base demonstrated cost-effectiveness in training with an F0.5 score of 69.00. The study also provides a benchmark for Filipino grammar checker models, contributing to the development of tools for low-resource languages.

Park et al. (2019) [24] developed an autonomous social robot companion, Tega, to enhance early literacy among preschool children, including English language learners. The system employed a model-free affective reinforcement learning framework, using multimodal affective cues such as facial expressions, body posture, and speech to dynamically adapt storytelling activities to each child's engagement level. In a three-month deployment with 67 children, the personalized interaction group demonstrated significantly higher engagement, vocabulary retention, and syntactic usage compared to non-personalized and baseline groups. This work demonstrated the potential of affect-driven personalization in improving early literacy outcomes for young learners.

The paper titled “Implementation of an AI English-Speaking Interactive Training System Using Multi-Model Neural Networks” [25] (Lu et al., published in IEEE Access on July 25, 2025) introduces an AI-based system AIESIT for enhancing users’ spoken English. The system integrates generative AI, speech recognition, and body posture and attention monitoring to create a realistic, interactive learning experience. A key feature is an AI agent whose mouth movements synchronize with spoken English, fostering immersion. The system utilizes speech recognition to evaluate spoken responses, OpenPoseNet to monitor posture and detect if learners stray or stand up, and an eye-based CNN to detect signs of drowsiness. It tracks learners’ performance by logging recognition accuracy and response time, providing a learning trajectory over time. User testing indicated that increased practice sessions via this system led to improved speaking ability and learner motivation.

Qiao and Zhao (2023) conducted a randomized controlled trial with 93 Chinese intermediate-level EFL learners to investigate the effects of AI-based instruction via the Duolingo app employing speech recognition, natural language processing, personalized feedback, and interactive exercises on L2 speaking skills and self-regulation [26]. The study assessed performance across multiple dimensions: fluency, vocabulary, grammatical accuracy, and pronunciation, as well as self-regulatory capacities. Findings showed that the AI-instructed group significantly outperformed the control group in all measured speaking skill areas and exhibited higher self-regulation which is considered as a pillar of Emotional Intelligence [26]. Additionally, speaking anxiety was found to negatively impact fluency and pronunciation, underscoring its disruption of specific aspects of oral performance [26].

Ruan et al. (2021) introduced EnglishBot [27], an AI-powered conversational system designed to enhance spoken English needs among second language learners. The chatbot engages users in interactive conversation on college-related topics and delivers adaptive feedback, simulating dynamic dialogue to mirror human conversation. The system reported improvements in learners’ fluency, achieving benchmarks akin to higher IELTS

speaking criteria, particularly in logical flow and expressive richness even when quantitative gains were modest. Importantly, EnglishBot offers a non-authoritative, low-pressure environment that helps mitigate learner anxiety and encourages practice without fear of judgment.

Yang et al. (2022) introduced "Ellie" [28], a task-based voice chatbot designed to function as an English conversation partner for Korean EFL learners aged 10–15. Deployed in classroom settings, Ellie engaged students in scenarios such as restaurant and shopping dialogues, encouraging substantial turn-taking averaging 9.6 turns per session using basic-level vocabulary (first 1,000-word band). Task success was high, with an 88.3% completion rate, indicating that Ellie's task prompts were appropriate and well-understood. Survey feedback from the learners reinforced Ellie's potential to enhance engagement and conversational practice in English-speaking classes, while also identifying areas for further refinement.

Buddy.ai is a voice-based AI tutor designed for young children (typically ages 3 to 8), offering engaging early-learning games and conversational practice in English. Children interact with Buddy, a friendly animated character, through voice, responding to flashcards, short questions, and playful dialogues that weave in vocabulary and language concepts across themes like animals, numbers, colors, and daily objects. Lessons are structured using the PPP (Presentation, Practice, Production) format and include features like spaced repetition, storytelling, and cartoon animations to reinforce learning [29].

1.2. Research Gap

Despite advances in RL-based dialogue management and automated grammar assessment, no existing work integrates these capabilities into a child-focused conversational agent that learns an adaptive dialogue policy from real-time speech-based grammar error analysis. Current educational agents either follow static scaffolding strategies or adapt

superficially based on preconfigured learner profiles [14], [15], without leveraging continuous grammar improvement as a reinforcement signal. Grammar Error Correction research treats the problem as an isolated text-editing task [16], [17], missing the pedagogical loop where error detection informs the next conversational move. Furthermore, the literature is heavily skewed toward deep RL methods for large-scale systems [13], [20], whereas a tabular Q-learning approach tailored to small, discrete state spaces could enable practical, data-efficient, and privacy-conscious adaptation in preschool or ESL settings. This presents a clear and feasible research opportunity to design and evaluate a speech-to-policy RL loop that personalizes interactions to a child's evolving grammar performance.

Despite growing research on explainable language assessment tools, current models rarely integrate this capability into real-time, voice-based, conversational learning environments for young ESL learners, nor do they provide actionable feedback dashboards tailored for educators and parents. The work of Ribeiro-Flucht et al. (2024) [21] confirms the feasibility of using linguistic complexity measures with explainable models but it remains focused on text-based assessments in Portuguese, lacking adaptation for speech-driven interaction, real-time grammar diagnostics, or personalized recommendations. Our proposed research addresses this gap by developing an XAI-powered dashboard that not only explains grammar issues detected in children's spoken responses, but also suggests targeted improvement strategies. This bridges the interpretability strengths demonstrated in the Portuguese context with the practical needs of Sri Lankan 4–6-year-old ESL learners, their caregivers, and educators.

While recent studies have advanced grammatical error detection and correction using transformer-based models, significant gaps remain for early childhood language learners. The work by "Enhancing Grammatical Error Detection using BERT with Cleaned Lang-8 Dataset" [22] demonstrates that rigorous dataset cleaning combined with fine-tuned BERT models can achieve high accuracy for adult learner English, yet it focuses exclusively on syntactic errors in written text from older learners and does not address the

unique linguistic characteristics of children's language. Similarly, "Balarila: Deep Learning for Semantic Grammar Error Correction in Low-Resource Settings" [23] targets semantic and syntactic errors in adult Filipino, relying on synthetic data generation due to limited resources, but it also does not consider the challenges posed by early second language acquisition in children, such as developmental grammar patterns, spoken language errors, and low literacy levels. Consequently, there remains a clear gap that there is a need of deep learning-based grammar correction system specifically tailored to young ESL learners in low-resource setting country like Sri Lanka, capable of addressing both semantic and syntactic errors in child-specific language use.

While Park et al. (2019) [24] showed that affective reinforcement learning can personalize literacy experiences and improve engagement, their focus was primarily on vocabulary and syntactic structures within a storytelling context using a physical robot. Current research has not adequately addressed real-time assessment and improvement of grammar accuracy, and conversational fluency for early second-language learners in resource-constrained settings by using simple conversations. Also, Park et al. used a physical robot introducing a gap for making systems more scalable and accessible especially in resource-constrained Sri Lankan contexts by removing the need for robotics infrastructure. Moreover, existing systems do not provide explainable, actionable feedback for teachers and parents. There is also limited exploration of connecting language development with Emotional Intelligence outcomes. These gaps highlight the need for an accessible, voice-based AI assistant that dynamically adapts dialogue complexity based on child performance, is fine-tuned for local speech patterns, and integrates transparent progress tracking with targeted learning recommendations.

While Lu et al.'s AIESIT [25] demonstrates promise combining multimodal AI to enhance spoken English in an interactive manner and tracking learner behavior, the system is primarily designed for professional or adult learners rather than early childhood or ESL learners in educational contexts. The system emphasizes speech recognition, response accuracy, posture, and engagement, but does not explicitly assess grammatical

correctness, or conversational complexity which are the key areas of improvement for ESL learners. Although it provides performance metrics, the system does not appear to dynamically adjust content difficulty based on learner performance. There's no reinforcement learning or adaptive dialogue complexity aligning with learners' evolving competence. The system offers performance tracking (recognition rate, response time), but does not offer explainable, actionable feedback for educators or parents, such as grammar-specific error corrections, progression insights, or pedagogical recommendations.

Despite demonstrating that AI-enhanced instruction can improve speaking skills and self-regulation among adult EFL learners, this study [26] reveals several gaps. This research focuses exclusively on older learners in a Chinese EFL context. There is a lack of evidence on how AI-based instruction impacts much younger learners especially early childhood ESL learners aged 4–6, whose linguistic development, cognitive capacities, and engagement patterns differ significantly. While the AI system provided feedback, it lacked dynamic personalization based on individual learner errors, grammar proficiency, or conversational fluency levels. It also did not adapt dialogue complexity over time through reinforcement learning or similar methods. The research measured speaking skills but did not include a mechanism to generate explainable insights or recommendations for educators or parents to support child-centered learning.

Buddy.ai [29] effectively supports vocabulary, pronunciation, and listening comprehension, it does not appear to evaluate grammar accuracy or facilitate conversational fluency through extended, adaptive dialogues. The app employs pre-designed learning sequences without clear mechanisms for real-time difficulty adjustment. It lacks reinforcement learning or performance-based adaptation to tailor interaction complexity based on the child's progress. Users receive weekly progress reports and stats, but these appear to focus on usage metrics rather than actionable, explainable insights.

Despite its advantages in providing adaptive speaking practice and reducing performance anxiety, EnglishBot has several limitations, particularly in relation to our proposed

system's goals. The system [27] targets older learners likely university-level students rather than early-childhood users (ages 4–6), missing the developmental needs of preschool ESL learners. EnglishBot [27] on fluency and expressive depth, but does not offer real-time grammatical correction in spoken input, leaving gaps in accuracy-focused learning. Also, the adaptive responses are oriented toward the learner, not designed to deliver interpretable insights for teachers or parents to understand performance.

While Ellie [28] showcases strong engagement and task alignment for adolescent EFL learners, several areas remain unexplored. Ellie targets older children (10–15 years) and does not address the developmental or linguistic needs of preschool-aged learners (4–6 years), who present distinct challenges and learning patterns. The study focuses on student-computer interaction metrics (e.g., turns per session, success rates), but does not offer detailed, interpretable feedback for teachers or parents that could inform targeted intervention or support. Ellie appears to reinforce basic vocabulary and dialogue fluency; it does not perform real-time corrective feedback on grammatical accuracy during speech. Although task design was effective, Ellie did not dynamically adapt dialogue complexity or difficulty based on individual learners' ongoing performance.

Table 1.1: Comparison of existing researchers with proposed system

	arXiv 2411.15523 (BERT for Grammatical Error Detection) [22]	Park et al., 2019 (Affective RL robot for early literacy) [24]	IEEE Access 2025 (AIESIT multi-model speaking trainer) [25]	Buddy.ai (English for kids) [29]	Proposed system
Age-specific for children	✗	✓	✗	✓	✓
Use of Explainable AI feedback for teachers/parents	✗	✗	✗	✗	✓
Developed for ESL learners	✓	✗	✓	✓	✓
Reinforcement learning / performance-based adaptation	✗	✓	✗	✗	✓
Real-time grammar correction in *speech*	✗	✗	✗	✗	✓
Emotional-Intelligence (EI) integration	✗	✓	✗	✗	✓

1.3. Research Problem

Language development during early childhood is a crucial determinant of later academic success, social integration, and emotional well-being. Research indicates that language acquisition in the early years is closely linked to the development of cognition, social cognition, and motor skills, influencing a child's ability to learn and interact effectively in various contexts [4], [8]. These skills are especially sensitive during the early "critical period" of language learning, where timely and appropriate exposure to linguistic input can significantly enhance language proficiency [5]. For Sri Lankan children aged 4–6 who are learning English as a second language, a lack of targeted, adaptive conversational practice opportunities may hinder both grammatical accuracy and overall communication competence.

Existing conversational agents have shown promise in enhancing language learning and learner motivation, but most systems are designed for adult learners or general language education contexts, lacking customization for preschool or early primary second-language learners [11], [18], [19]. Furthermore, most grammar correction technologies and dialogue systems are trained on adult speech, which do not adequately reflect the unique grammatical structures and error patterns found in children's spoken language [16], [17], [21].

A key gap lies in creating a system that combines accurate speech recognition for children's voices, grammar error detection and correction tailored to early learners, and adaptive, personalized conversation management. Training speech recognition models solely on child audio presents significant challenges due to scarcity of datasets and high transcription error rates, which can negatively impact downstream processing and grammar analysis [22], [23]. Similarly, while reinforcement learning has been successfully applied in dialogue systems to optimize responses and personalize learning experiences [13], [24], there is limited research on leveraging it to adapt conversational

agents to a child's grammatical level and progressively scaffold their language use in a natural, spoken format.

Given the established link between language skills and emotional intelligence pillars such as social skills and communication [4], [8], and the proven motivational effects of interactive, adaptive language learning tools [18], [25], there is a pressing need for a system designed specifically for young second-language learners. Such a system should enable teachers and parents to monitor progress, understand a child's strengths and weaknesses, and receive explainable AI–based recommendations for targeted improvement.

2. OBJECTIVES

2.1. Main Objectives

The main objective of this research is to help parents and teachers to monitor child's emotional intelligence development and find whether any critical situations has occurred.

2.2. Specific Objectives

The main objective of this research component is to develop and implement a feature for a web-based application for identification of English Language development skills in children by teachers and parents. This is a conversational voice assistant, specifically tailored for Sri Lankan children aged 4–6 learning English as a second language, that can accurately recognize speech, detect, and correct grammatical errors, and adapt conversation prompts using reinforcement learning to match each child's language proficiency level. The system will provide teachers and parents with real- time progress tracking, detailed insights into the child's language development, and explainable AI–based recommendations to improve grammar, conversational ability, and overall communication skills.

- Speech Recognition Integration

Integrate an existing fine-tuned speech recognition model from Hugging Face to accurately transcribe spoken English from young children, ensuring reliable input for grammar analysis.

- Grammar Error Detection and Correction

Fine-tune an existing grammar correction model using a custom-created text dataset that reflects the grammatical patterns of Sri Lankan second-language learners aged 4–6. (e.g., incorrect verb forms, missing articles, subject-verb agreement).

- Adaptive Conversational Design

Implement a reinforcement learning–based mechanism to adapt conversation prompts and difficulty levels according to each child's language proficiency and observed grammar error trends.

- Teacher & Parent Dashboard

Develop a web-based dashboard enabling teachers and parents to monitor individual child performance, track progress, EI tracking and receive explainable AI–based recommendations for targeted language improvement.

3. METHODOLOGY

3.1. Overall System Methodology

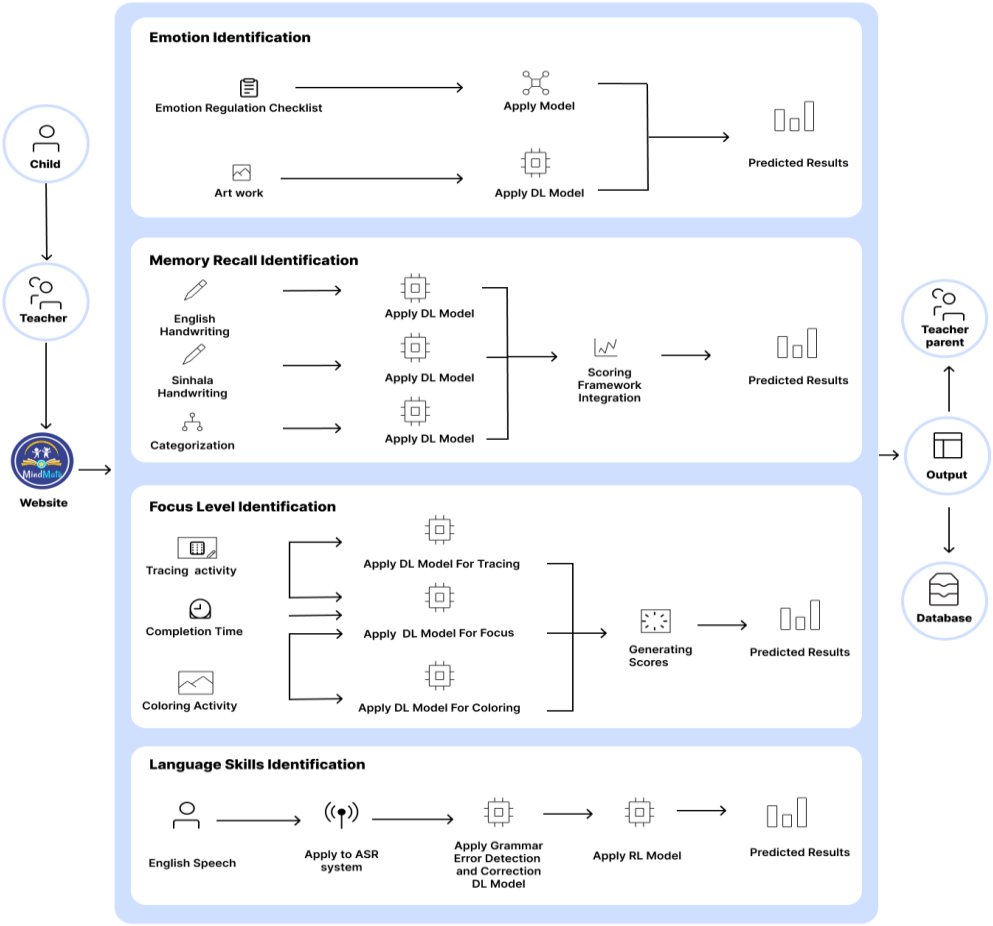


Figure 3.1: High Level System Overview Diagram

As illustrated in Figure 3.1, the system overview provides a comprehensive representation of a complete overview of the structure of the entire system. The system leverages advanced technologies, including image processing, audio processing, text processing, RL, ML, and multi-model architectures, to analyze students' performance through

cognitive skills, motor skills, emotions and language development. The system operates by collecting data through user interactions, where students engage with conversational voice assistants and image inputs of children's handwriting, drawings, behavioral activities and line activities captured by teachers for analysis. The system then processes this data, extracting relevant features using sophisticated algorithms. A multimodal architecture is employed to integrate these features and make predictions regarding the student's performance across various areas such as emotion, focus level, procedural memory and language skill which can assess Emotional Intelligence Pillars such as self-awareness, self-regulation, social skills, motivation and empathy. Feedback is subsequently provided based on the analysis in a teacher parent dashboard, which keeps the parent and teacher connected. The system leverages REST API Application Program Interfaces to facilitate seamless communication between different components, ensuring efficient data flow and real-time feedback. This holistic approach enables to get personalized feedback of children for teachers and parents.

3.2. Component Methodology

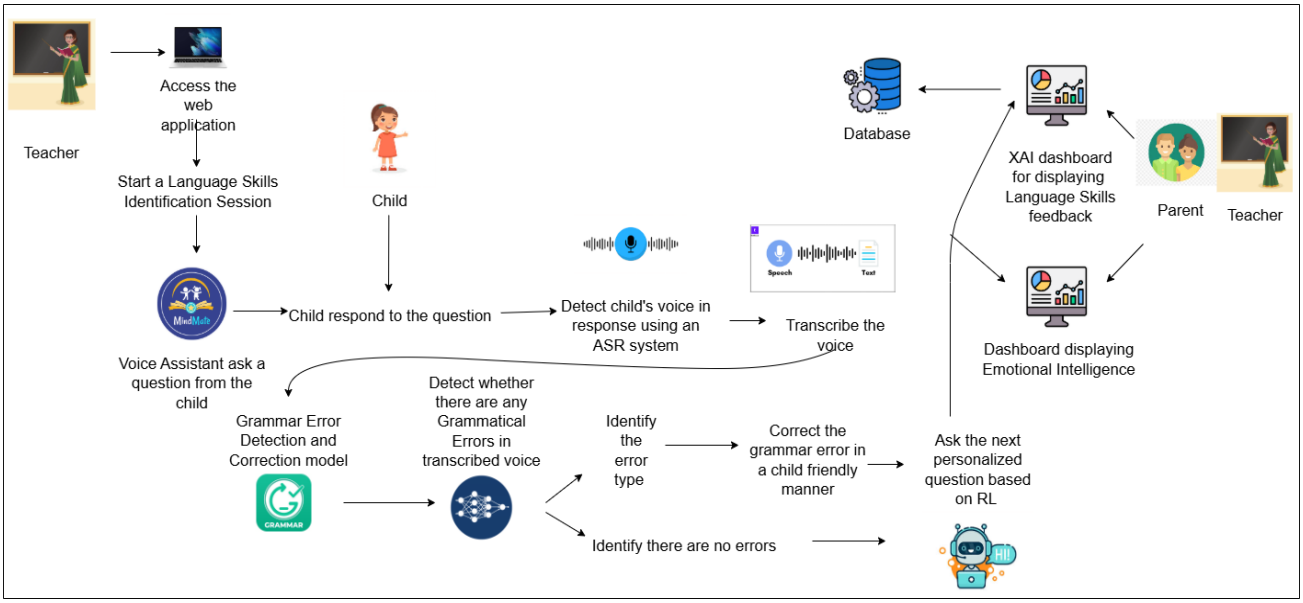


Figure 3.2: High Level Component Diagram

The proposed system, as shown in Figure 3.2 will be developed using a design, development, and evaluation approach, integrating speech recognition, grammar correction, reinforcement learning, and a teacher/parent monitoring dashboard into a single web-based platform. Since training an automatic speech recognition (ASR) model from scratch on child audio data is challenging due to dataset scarcity and the high error rates that would negatively impact downstream grammar analysis [30], this research will integrate an already fine-tuned ASR model from Hugging Face to ensure reliable transcription of children's speech. The transcribed text will then be processed through a grammar error detection and correction module, created by fine-tuning an existing grammar correction model using a custom dataset that reflects the grammatical patterns and common errors of Sri Lankan second-language learners aged 4–6. This is necessary because most existing grammar correction systems are trained on adult-written corpora and fail to capture child-specific linguistic patterns [17]. To make the learning experience adaptive, reinforcement learning will be employed to personalize conversation prompts according to each child's language proficiency and error trends, building on evidence that reinforcement learning can enhance adaptability and engagement in dialogue systems [13]. The conversation design will involve short, age-appropriate, five-minute sessions where the voice assistant asks simple, open-ended questions to elicit natural speech, aligning with research showing that conversational agents improve learner motivation and attitudes toward language learning [11], [18], [19]. All interaction data will be presented in a web-based dashboard for teachers and parents, providing grammatical performance summaries, progress tracking, and explainable AI–based recommendations, supporting informed instructional decisions. The system will be evaluated through pilot testing with Sri Lankan ESL learners, measuring speech recognition accuracy, grammar correction performance, and reinforcement learning adaptation rates, alongside qualitative teacher, and parent feedback to assess usability and perceived effectiveness.

3.2.1. Requirement Gathering

The requirements for our research were primarily gathered through a series of frequent, structured consultations with our external supervisor, Dr. Naren Deepan, a Senior Lecturer from the School of Psychology at Sri Lanka Institute of Information Technology. This direct engagement was crucial for bridging the gap between the technical aspects of an AI-based system and the nuanced domain of child psychology. We gained a comprehensive understanding of the psychological constructs of emotional intelligence in early childhood, including key indicators, developmental milestones, and age-appropriate emotional expressions. This information was vital for defining the core functionalities of the AI model.

Also, series of consultations with preschool and school teachers was conducted to gain deep domain knowledge from an educational perspective. These discussions provided invaluable insights into the specific challenges faced by Sri Lankan children aged 4–6 who are learning English as a second language. Teachers highlighted common pronunciation difficulties, grammatical errors that stem from native language syntax, and the need for a system that encourages conversational practice in a low-anxiety environment. Their feedback helped us to define core functionalities such as accurate speech recognition, grammar detection, and the development of an adaptive conversational engine.

Further, a comprehensive survey was conducted to validate these findings and to highlight the broader demand for such a system. The survey results underscored the high level of interest among parents and educators in an application that can provide detailed insights into language development.

3.2.2. Data Gathering

For this component, a custom text dataset will be created using common grammatical errors made by ESL learners in this age group to train the grammar error detection and correction model. Also, real interaction logs will be collected using a rule based virtual student.

3.3. Research Constraints

3.3.1. Data Collection Challenges

The primary data collection challenge is the creation of a high-quality, annotated dataset of grammatical errors specific to children. This is a time-consuming process.

Collecting authentic speech data from preschool children presented significant challenges. Many preschools declined permission to record children, and those that allowed participation typically permitted recordings of only one or two children, with limited time allocations of around one hour. This made it difficult to collect a sufficient volume of audio data for training while also limiting variability, since gathering many samples from a small number of children does not capture the diversity of speech patterns needed for robust modeling. To overcome this, publicly available speech datasets were explored. However, most were not specific to children in the 4–6 age range and were often unannotated. Given these constraints, the decision was made to utilize an already fine-tuned speech recognition model available on Hugging Face, rather than attempting to train a model from scratch under insufficient data conditions.

3.3.2. Accuracy of Models

The grammar correction model's accuracy will be directly proportional to the quality and diversity of the custom dataset. If the dataset does not cover a wide range of error types, the model may struggle to generalize to unseen errors.

Although a fine-tuned speech recognition model from Hugging Face will be used, children’s voices especially at younger ages can still lead to higher error rates due to pronunciation variability, disfluencies, and background noise.

3.3.3. Ethical Concerns

Working with children requires strict adherence to ethical standards. Parental consent and child assent must be obtained before participation.

The fine-tuned model could potentially develop biases if the training data is not diverse enough. Need to address this issue which could lead to the model performing better for certain demographics or language patterns than for others, creating an unequal experience.

3.3.4. Resource Constraints

Computational Resources

Fine-tuning a large language model is a computationally intensive process requiring access to powerful GPUs. This limits the number of training iterations and model architectures that can be explored within the project timeline.

Developing and hosting a web application with integrated ASR, grammar correction, reinforcement learning, and dashboard features requires significant computational resources. Although cloud services can be leveraged, budget constraints may limit prolonged use of high-performance computing instances for large-scale testing or fine-tuning.

3.3.5. Human Resources

Limited access to experts in both educational psychology and information technology.

3.3.6. Time Constraints

Collecting data, fine-tuning the grammar correction model, developing the reinforcement learning mechanism, integrating the dashboard, and conducting pilot testing must all be completed within a relatively short research period, requiring careful time management. Creating a high-quality, annotated dataset of a sufficient size is the most time-consuming part of this project.

The iterative process of fine-tuning the model, evaluating its performance, and making adjustments is also subject to time constraints. A limited timeline may restrict the ability to explore multiple model configurations or fine-tune for extended periods to achieve optimal performance.

3.3.7. Other Challenges

Adaptability: Designing RL models that can adapt to different student ability levels, and diverse educational environments.

User Engagement: Ensure that the system is user-friendly, accessible, and beneficial in real-world educational settings to be used by teachers and students.

4. PROJECT REQUIREMENTS

4.1. Functional Requirements

Authentication & Access Control

- Teachers and parents must be able to log in with child-specific credentials.
- Role-based access control: teachers can initiate sessions, parents can only view progress.

Language Skill Assessment Session

- Voice assistant must present questions in English appropriate for 4–6-year-old learners.
- Child must be able to respond verbally. Responses are transcribed via ASR (Automatic Speech Recognition).
- System must detect and highlight grammar errors in real time.
- Provide child-friendly immediate feedback.

Dashboard & Progress Tracking

- Teachers' and Parents' dashboard: Real-time session monitoring, session summary (error types, accuracy, progress) and Emotional intelligence insights.

Data Storage and Privacy

- Store all session data securely in a database.
- Support progress tracking longitudinally.
- Parents can only access data linked to their child.

4.2. User Requirements

Teacher Requirements

- Ability to log in securely using child-specific credentials.

- Should start a language skill assessment session with the conversational voice assistant and have the child interact naturally with the voice assistant via speech-based questions and answers.
- Should see real-time grammar error detection in the child's responses and session summary in a dashboard.
- Should be able to help the students to improve language skills using the personalized insights given by the system.

Parent Requirements

- Ability to log in securely using child-specific credentials.
- Should be able to monitor their child's progress and help to improve language skills using the personalized insights given by the system.

Child Requirements

- Interact with the voice assistant in a child-friendly, engaging, and safe environment.
- Experience age-appropriate questions and feedback.

System Administrator Requirements

- Manage user accounts.
- Ensure data security and privacy of child data.

4.3. System Requirements

Hardware

- Microphone & speakers/headset for classroom sessions or else laptop can be used.

Software

- Web app (React-based) accessible via browser.

4.4. Non-functional Requirements

1. Performance

- Real-time speech-to-text transcription and grammar error detection should happen with minimal delay.
- The system must handle at least 10–20 concurrent sessions without significant lag (scalable for multiple classrooms).

2. Usability

- Child-friendly voice assistant interface with simple prompts and encouragement.
- Dashboards must be intuitive and visually clear for teachers and parents.

3. Reliability & Availability

- System uptime should be high during school hours.
- The application must provide consistent and accurate results.

4. Scalability

- Support for scaling from a few children (pilot) to thousands of children across multiple preschools.

5. Maintainability

- The application should be designed modular so ASR, grammar correction, and dashboard can be updated independently.

4.5. Technical Requirements

1. Automatic Speech Recognition (ASR)

Available Technologies

- Google Speech-to-Text API has features such as high accuracy, easy integration and, multilingual. But commercial cost per API call is high and, child speech accuracy is lower without customization.

- Microsoft Azure Cognitive Services (ASR) has features such as enterprise-ready, scalable and, supports custom acoustic models. But this requires heavy customization for children's voices.
- OpenAI Whisper (open-source) has features such as state-of-the-art, robust to accents/noisy environments and, open-source. But heavy computational cost, requires GPU acceleration for real-time use are some disadvantages.
- Hugging Face Pretrained Models like Wav2Vec2, Whisper fine-tuned on children's speech has features such as fine-tuning possible on domain-specific data and, strong open-source community support. But this requires dataset and infrastructure for fine-tuning.

Selected Technology:

Whisper / Hugging Face pretrained ASR because it is open-source, robust in noisy preschool environments, supports fine-tuning on limited child speech data. More suitable than commercial APIs which are costly and less adaptable to child-specific patterns.

2. Grammar Error Detection and Correction (GEC)

Available Technologies

- Rule-based grammar checkers like LanguageTool are lightweight and, interpretable. But poor at handling spoken child language errors, misses context.
- Transformer-based GEC Models like GECToR, T5, BART, mT5 has features such as state-of-the-art, context-aware and, capable of handling spoken language errors. But this require training data tailored to child speech.
- Commercial Tools like Grammarly, ProWritingAid APIs has high accuracy for written English. But those are proprietary, optimized for adult writing and, not child speech.

Selected Technology:

Transformer-based GEC model (T5/BART fine-tuned on child-like spoken text) is mostly suitable because unlike Grammarly or rule-based tools, fine-tuned transformer models can adapt to the grammatical development stage of children (e.g., missing articles, verb conjugations). This ensures higher relevance and pedagogical value.

3. Conversational Agent & Reinforcement Learning

Available Technologies

- Dialogflow (Google), Rasa (Open-source), IBM Watson Assistant are prebuilt NLP pipelines which can be setup quickly. But limited adaptability to grammar-based personalization.
- Custom RL-driven Dialogue System (Python RL frameworks: Stable-Baselines3, RLlib) has features such as adaptive learning and, ability to adjust difficulty/questions based on child's performance. But this requires model training and ongoing tuning.

Selected Technology:

Custom RL-driven conversational agent using an LLM for dialogue flow and RL (Stable-Baselines3/Rlib) for personalization which combines the structured conversational capabilities of with adaptive reinforcement learning for real-time personalization, which commercial agents cannot provide.

4. Dashboard and User Interface

Available Technologies

- Frontend**:** Angular, React, Vue.js

  React chosen due to wide adoption, component reusability, and ecosystem support.

- Backend**:** Flask/Django (Python) vs Spring Boot (Java)

Python Flask chosen for rapid integration with ASR & NLP models.

- Database**:** PostgreSQL vs MySQL vs MongoDB

  PostgreSQL chosen for structured relational data (users, sessions, grammar stats).

Selected Technology:

Stack:
Frontend: React.js, Backend: Flask (Python), Database: PostgreSQL. Python backend aligns with NLP/ML pipeline, React provides smooth dashboards, PostgreSQL ensures reliable structured storage.

5. Testing

Software Testing

- Unit Testing: Unit testing is done by each member of the team for each individual component APIs (ASR endpoint, GEC correction API, Dashboard components).
- Integration Testing: End-to-end pipeline of each component is tested (Speech → ASR → GEC → Feedback → Dashboard) and all the four components will be integrated.
- System Testing: The overall system would be tested to check the overall functions of the application. User role testing (teacher, parent, child).
- Usability Testing: Conduct evaluations with teachers and parents for dashboard clarity.

Table 4.1: Tools and technologies

Programming Languages	Python Flask, React
Algorithms	ML, RL, Deep learning
Database	PostgreSQL

4.6. Expected Test Cases

Table 4.2: Expected test cases

Test ID	Description	Precondition	Test steps	Expected Results
TC001	User registration	User opens the application	1. User navigate to registration page 2. User creates a username and a password 3. Click registration button	User account is successfully created and stored in the database, User directs to login page
TC002	Login with invalid credentials	User has a registered account in the database	1. Enter invalid username/password 2. Click login button	Display an error message as “Invalid credentials”
TC003	Start session for a child	User has a registered account in the database	1. Login to the child account 2. Navigate to language skills section 3. Click start a session button 4.Give the device to the child	Conversation session begins
TC004	Grammar errors displayed in real-time	User has an active session	1. Child responds with “She go to school”	System detects the error and correct it.
TC005	Correct grammar response	User has an active session	1. Child responds with “She goes to school”	System continues the conversation
TC006	View session summary	Session completed	1. Click on view analysis button	System shows the currently completed session analytics
TC007	Parent sees emotional intelligence insights	Parent log into the system using child account	1. Navigate to dashboard page 2. Explore emotional intelligence skills	System display the overall emotional intelligence of that specific child

5. COMMERCIALIZATION

5.1. Customer value

Through the survey results we found out, 85% of the people would like to see an AI tool to monitor the emotional intelligence. They also highlighted how important emotional intelligence monitoring is for the early childhood development. The target audience for this application will be parents and teachers. Considering these results below are some of the steps that will be considered to commercialize the application.

5.2. Competitive analysis

Similar applications on emotional intelligence without AI models has an average price range of $9.99 monthly. Applications that use AI models has an average price of $20 monthly subscription.

5.3. Comparison of proposed application with existing application

Exiting applications	Our application
Gamified applications that increase digital addictions of the children	Uploading the work the child has done to reduce digital addiction
Facial expression capturing to identify emotions	Does not capture or store children's facial expressions or biometric identifiers
Use activities to improve child specific skill area.	Use activities not only to improve their skills but also to get insights on emotional intelligence based on those activities.
Limited parental involvement or communication	Facilitates and enhances parent-teacher interaction and collaboration
Focus more on cognitive, thinking, and emotions with reward-based motivation	Provides meaningful insights and personalized recommendations to parents and teachers based on the child's activity outputs

Primarily rely on digital input (typing, drag-and-drop, touch-based tasks)	Ensures that no direct digital input is involved in children's activities.
Generic design without consideration for cultural or linguistic diversity	Incorporates both Sinhala and English, ensuring cultural and linguistic appropriateness.

5.4. User benefits of our applications

- Monitor child's emotional intelligence using multiple methods
- Builds emotional intelligence at an early age, which supports learning, social skills, and resilience.
- Helps teachers identify at-risk students early (e.g., children showing signs of sadness, anger, withdrawal)
- Teachers and parents both can help children to improve suggested activity skills based on dashboard feedback.

5.5. Cost analysis

On average a web hosting platform will cost $7 with PostgreSQL database support (eg: Hostinger). This will be the rough hosting cost of our application.

5.6. Pricing plan

Considering the above factors this application will have a one month free trial period, which will be used to gain customers into the application. After the one month trial, the user will have to pay $15 monthly subscription.

Standard plan ($15) – One subscription per student.

Family plan ($30) – One subscription per three students.

School plan ($40) – Class registration ability, one subscription per 20 students.

6. GANTT CHART AND WORK BREAKDOWN STRUCTURE

6.1. Gantt chart

Table 6.1: Gantt Chart

Task Name	June	July	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May
Research topic selection												
Topic Assessment												
Project Charter												
Study on research area												
Project Proposal Report												
System designing and planning												
Implementation of functions												
Integration level 1												
Testing level 1												
Progress presentation 1												
Check list 1												
Prepare research paper												
Research paper												
Implementation of functions												
Testing level 2												
Progress presentation 2												
Check list 2												
Final report												
Final presentation												

6.2. Work Breakdown Structure

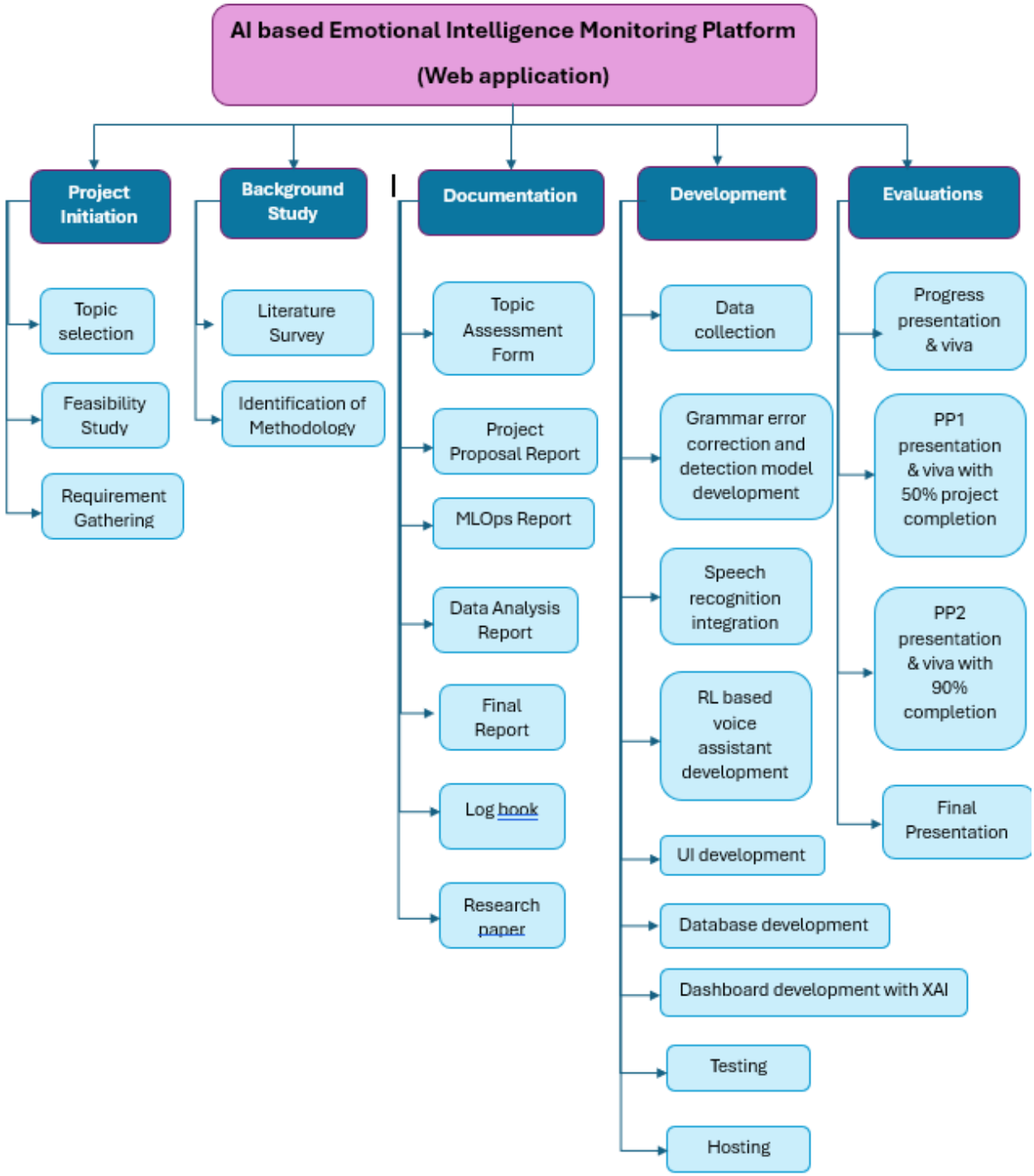


Figure 6.1: Work breakdown structure

REFERENCES

[1] V. Pawar, A. Vhatkar, P. Chavan, S. Gawankar, and S. Nair, “The future of emotional engineering: Integrating generative AI and emotional intelligence,” in Proc. 2024 8th Int. Conf. Computing, Communication, Control and Automation (ICCUBEA), Pune, India, Aug. 2024, pp. 1–6, doi: 10.1109/ICCUBEA61740.2024.10775105.

[2] I. Coronado-Maldonado and M.-D. Benítez-Márquez, “Emotional intelligence, leadership, and work teams: A hybrid literature review,” Heliyon, vol. 9, no. 10, art. e20356, Oct. 2023, doi: 10.1016/j.heliyon.2023.e20356.

[3] J. Ebert et al., "Teaching Emotion and Creativity Skills to Preschoolers Through a Child Workshop," The International Journal of Creativity & Problem Solving, vol. 25, no. 2, pp. 23–35, 2015.

[4] "Impact of language development on psychosocial and socioemotional well-being," Encycl. Early Child. Dev., [Online]. Available: https://www.child-encyclopedia.com/language-development-and-literacy/according-experts/impact-language-development-psychosocial-and.

[5] "Critical period hypothesis," in Wikipedia, The Free Encyclopedia, [Online]. Available: https://en.wikipedia.org/wiki/Critical_period_hypothesis.

[6] C. Samarajeewa and L. A. Mohammed, "Obstacles in second language acquisition: Linguistic, psychological, social, cultural, and pedagogical challenges with a focus on secondary ESL Education in Sri Lanka," World J. Adv. Res. Rev., vol. 25, no. 01, pp. 087-094, 2025, doi: 10.30574/wjarr.2025.25.1.3582.

[7] R. R. Divekar, H. Lepp, P. Chopade, A. Albin, D. Brenner, and V. Ramanarayanan, "Conversational Agents in Language Education: Where They Fit and Their Research Challenges," presented at the IUI Workshop on User Interfaces for Conversational Agents, 2021. [Online]. Available: https://vikramr.com/pubs/Divekar_IUI_CUI_Workshop_2021.pdf.

[8] M. E. S. Sulistyawati, "The Effects of Emotional Intelligence on Students' Speaking Skills," EDULANGUE, vol. 1, no. 1, pp. 39-66, Aug. 2018, doi: 10.20414/edulangue.v1i1.195.

[9] T. Andrienko, N. Chumak, and V. Genin, “Emotional Intelligence and Acquisition of English Language Oral Communication Skills,” Advanced Education, vol. 7, no. 15, pp. 66–73, Jul. 2020, doi: 10.20535/2410-8286.201013.

[10] N. Mulvey and L. Jenkins, “Language Skills as Predictors of Social Skills and Behaviors in Preschool Children,” Contemporary School Psychology, vol. 25, pp. 503–514, Apr. 2020, doi: 10.1007/s40688-020-00281-1.

[11] M. Warschauer, F. Xiao, P. Zhao, H. Sha, and D. Yang, "Conversational agents in language learning," J. China Comput. Assist. Lang. Learn., vol. 4, no. 2, pp. 300-325, 2024. doi: 10.1515/jccall-2023-0002. [Online]. Available: https://www.researchgate.net/publication/369605653_Conversational_agents_in_language_learning.

[12] "Conversational agents in language learning," LearnIt2Teach, [Online]. Available: https://bib.learnit2teach.ca/blog/conversational-agents-in-language-learning/.

[13] T. Kwan, A. Celikyilmaz, and Y. Chen, “A Survey of Reinforcement Learning for Task-Oriented Dialogue Systems,” ACM Comput. Surv., vol. 55, no. 6, pp. 1–38, Dec. 2023.

[14] M. D’mello and S. Graesser, “Dialog-based intelligent tutoring systems: A review of recent advances,” AI in Education, vol. 29, no. 2, pp. 1–20, 2021.

[15] F. Winkler, M. Söllner, and J. M. Leimeister, “Designing Chatbots for Education: A Systematic Literature Review,” Educational Technology & Society, vol. 26, no. 1, pp. 42–59, 2023.

[16] D. M. Nguyen, T. Nguyen, and N. H. Pham, “Reinforcement Learning for Grammatical Error Correction,” in Proc. 55th Annu. Meeting Assoc. Comput. Linguist., 2017, pp. 326–331.

[17] C. Chollampatt and H. T. Ng, “Neural Quality Estimation for Grammatical Error Correction,” in Proc. Conf. Empirical Methods in Natural Language Processing, 2018, pp. 2528–2539.

[18] L. F. Chen, H. Huang, and Y. H. Chang, “The effects of a chatbot-based language learning system on learners’ motivation and attitudes,” Computers & Education, vol. 166, p. 104–150, 2021.

[19] M. Jia et al., “Conversational agents in language education: A scoping review,” Computer Assisted Language Learning, vol. 37, no. 1–2, pp. 87–113, 2023.

[20] P. Ezen-Can and K. Boyer, “Unsupervised Classification of Student Dialogue Acts with Latent Dirichlet Allocation,” Educational Data Mining, pp. 20–29, 2015.

[21] H. Yang, Y. Ma, J. Lin, J. Chen, and Y. Lin, "Assessing the Quality of AI-generated Conversations for Language Learning," presented at the NAACL-HLT Conference, 2024. [Online]. Available: https://aclanthology.org/2024.bea-1.17.pdf.

[22] K. K. Lee, W. T. Huang, T. Y. Wu, and C. H. Lai, "Learning a Second Language via AI-Generated Conversational Agents: A Survey," arXiv, [Online]. Available: https://arxiv.org/html/2411.15523v1.

[23] P. Espiritu, J. Jadie, A. Ponce, and C. Cheng, “Balarila: Deep Learning for Semantic Grammar Error Correction in Low-Resource Settings,” in Proc. First Workshop in South East Asian Language Processing (SEALP), Nusa Dua, Bali, Indonesia, Nov. 2023, pp. 21–29, doi: 10.18653/v1/2023.sealp-1.3.

[24] H. W. Park, I. Grover, S. Spaulding, L. Gomez, and C. Breazeal, “A Model-Free Affective Reinforcement Learning Approach to Personalization of an Autonomous

Social Robot Companion for Early Literacy Education," in Proc. AAAI Conf. Artificial Intelligence, vol. 33, no. 1, pp. 687–694, Jul. 2019, doi: 10.1609/aaai.v33i01.3301687.

[25] C. -T. Lu, Y. -Y. Lu, Y. -R. Lu, Y. -C. Pan and Y. -C. Liu, "Implementation of an AI English-Speaking Interactive Training System Using Multi-Model Neural Networks," in IEEE Access, vol. 13, pp. 132052-132066, 2025, doi: 10.1109/ACCESS.2025.3592632.

[26] H. Qiao and A. Zhao, "Artificial intelligence-based language learning: illuminating the impact on speaking skills and self-regulation in Chinese EFL context," Frontiers in Psychology, vol. –, no. –, p. –, 2023, doi: 10.3389/fpsyg.2023.1255594.

[27] S. Ruan, L. Jiang, Q. Xu, Z. Liu, G. Davis, E. Brunskill, and J. A. Landay, "EnglishBot: An AI-Powered Conversational System for Second Language Learning," in Proc. IUI, 2021, pp. 434–444, doi: 10.1145/3397481.3450648.

[28] H. Yang, H. Kim, J. H. Lee, and D. Shin, "Implementation of an AI Chatbot as an English Conversation Partner in EFL Speaking Classes," ReCALL, vol. 34, no. 3, pp. 327–343, Sep. 2022, doi: 10.1017/S095834402200008X.

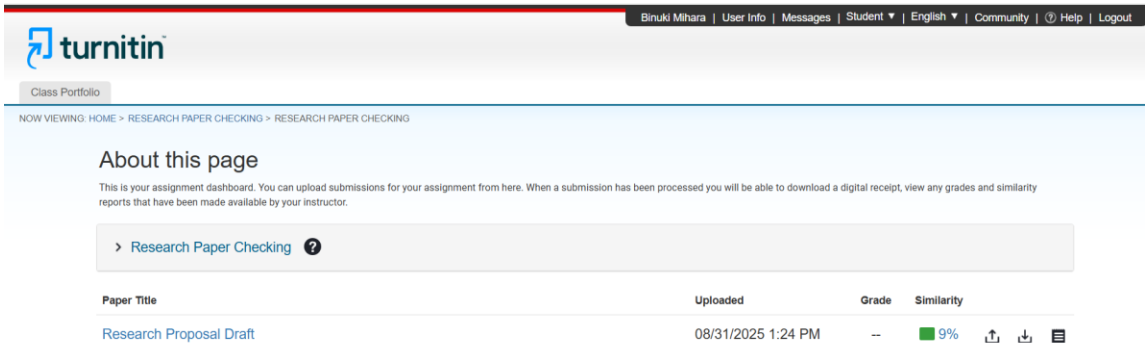
[29] Buddy.ai, "English for Kids." [Online]. Available: https://buddy.ai/en/english-for-kids. [Accessed: Aug. 26, 2025].

[30] V. Bhardwaj et al., "Automatic Speech Recognition (ASR) Systems for Children: A Systematic Literature Review," Appl. Sci., vol. 12, no. 9, Art. no. 4419, 2022, doi: 10.3390/app12094419.

APPENDICES

Appendix A: Online Survey

https://forms.office.com/r/BVupx17rcQ

Appendix B: Logo

Appendix C: Plagiarism Score using Turnitin